

RAJ KAPADIA

Senior Dialogflow AI Architect | Conversational AI Expert | 150+ Chatbots Delivered

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[GitHub](#)

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Results-driven Conversational AI Architect with **7+ years** of hands-on experience specializing in **Google Dialogflow ES, CX, and Conversational Agents (DFCX)**. Proven expertise in designing and deploying enterprise-grade chatbot and voicebot solutions serving millions of users across multiple industries. Successfully delivered **150+ production chatbots** with demonstrated ability to lead cross-functional teams, architect scalable NLU systems, and integrate cutting-edge LLM technologies. Strong background in cloud infrastructure (GCP/AWS), full-stack development, and AI/ML model deployment. Active community contributor through YouTube tutorials and open-source projects.

PROFESSIONAL EXPERIENCE

Team Lead – AI & Machine Learning

Sep 2023 – Apr 2025

Let's Enkindle

📍 Gujarat, India

Led enterprise AI initiatives, managing a team of 5 developers to deliver production-grade conversational AI and ML solutions for diverse client requirements.

Core Responsibilities:

- ▶ **Technical Leadership:** Directed end-to-end development lifecycle for conversational AI projects using **Dialogflow ES, CX, and Conversational Agents**; conducted architecture reviews, sprint planning, and technical mentorship sessions
- ▶ **Voice & IVR Solutions:** Designed enterprise voice bot architectures integrating with **Twilio, Genesys, and Avaya** telephony systems; implemented DTMF handling, call transfer, and barge-in capabilities
- ▶ **NLU Design & Optimization:** Created comprehensive intent taxonomies, entity extraction strategies, and conversation flow designs; achieved **92% intent accuracy** through iterative training and testing
- ▶ **Webhook Development:** Architected robust fulfillment backends using **Python + FastAPI** and **NodeJS + Express**; implemented error handling, logging, and monitoring with Cloud Logging
- ▶ **CI/CD Implementation:** Established automated deployment pipelines using **Cloud Build and Terraform** for Dialogflow agent versioning, backup, and multi-environment promotion
- ▶ **Stakeholder Collaboration:** Translated complex business requirements into technical specifications; conducted client demos, UAT sessions, and knowledge transfer workshops
- ▶ **Quality Assurance:** Implemented conversation testing frameworks, A/B testing strategies, and analytics dashboards using **BigQuery and Looker Studio**

Key Achievements:

- ★ **Text-to-SQL WhatsApp Bot:** Built LLM-powered natural language interface enabling business users to query databases conversationally; integrated with **WhatsApp Business API**; reduced reporting dependency on IT by **60%**
- ★ **LLM Fine-Tuning Pipeline:** Developed scalable infrastructure for domain-specific model fine-tuning; implemented efficient inference serving achieving **40% latency reduction**
- ★ **Visual Product Search:** Architected deep learning-based reverse image search for tile manufacturer processing **100,000+ SKUs**; increased product discovery by **35%**

Lead AI & ML Developer

Jan 2022 – Aug 2023

The Other Fruit / Nes Tech Ltd

📍 United Kingdom (Remote)

Spearheaded AI/ML product development for UK-based technology company, leading a team of 3 developers in building innovative machine learning solutions.

Core Responsibilities:

- ▶ **ML System Architecture:** Designed end-to-end machine learning pipelines from data collection through model deployment; implemented **MLOps practices** including model versioning and A/B testing
- ▶ **API Development:** Built production-grade **RESTful APIs** using **FastAPI** with async processing, request validation, authentication, and comprehensive API documentation (OpenAPI/Swagger)
- ▶ **Data Pipeline Engineering:** Developed automated ETL pipelines for image scraping, preprocessing, and augmentation; scheduled jobs using **Cloud Scheduler and cron**
- ▶ **Vector Search Implementation:** Deployed **Elasticsearch** clusters for scalable similarity search with custom scoring algorithms and faceted filtering capabilities
- ▶ **Infrastructure Management:** Managed **AWS infrastructure** (EC2, S3, Lambda, RDS, SageMaker) with Infrastructure-as-Code using CloudFormation
- ▶ **Documentation & Training:** Created comprehensive technical documentation, runbooks, and conducted knowledge transfer sessions for client engineering teams

Key Achievements:

- ★ **Enterprise Image Search:** Built visual similarity engine using **TensorFlow and Elasticsearch**; automated image collection from multiple web sources with comparative analysis
- ★ **Biometric Authentication:** Designed face recognition system using **dlib** achieving **99.2% accuracy**; implemented voice-as-password 2FA for enhanced security
- ★ **Content Moderation:** Built toxic comment classifier using **PyTorch** with **94% accuracy**; deployed for real-time filtering across platform
- ★ **SaaS Platform:** Developed full-stack applications using **NextJS, TailwindCSS, ShadCN, Drizzle ORM, and PostgreSQL**

Independent Dialogflow Consultant

2018 – Present

Fiverr & Upwork – Top-Rated Seller

Remote / Global

Delivered 150+ conversational AI projects for enterprise clients across retail, healthcare, finance, and telecommunications sectors with 98% client satisfaction.

Core Responsibilities:

- ▶ **End-to-End Chatbot Development:** Designed, developed, and deployed **Dialogflow ES/CX chatbots** from requirements gathering through production launch and maintenance
- ▶ **Multi-Channel Integration:** Implemented chatbots across **WhatsApp Business API, Facebook Messenger, Telegram, Slack, web widgets**, and custom mobile apps
- ▶ **Voice Bot Implementation:** Built IVR solutions with **Twilio, Vonage, and AudioCodes** integration; implemented SSML for natural speech synthesis
- ▶ **CRM Integration:** Connected chatbots with **Salesforce, HubSpot, Zendesk, Freshdesk**, and custom CRM systems for seamless ticket creation and customer data sync
- ▶ **E-commerce Solutions:** Developed conversational commerce bots with product catalog browsing, cart management, and **payment gateway integration** (Stripe, Razorpay)
- ▶ **AI Agent Development:** Created autonomous agents using **OpenAI Agent SDK and Google ADK** for complex multi-step task execution
- ▶ **RAG Implementation:** Built knowledge-augmented chatbots using **vector databases** (Pinecone, Weaviate, Qdrant) for accurate domain-specific responses

Assistant Professor – Electrical Engineering

Aug 2013 – Jan 2022

Govt. Engineering College Modasa & SPCE Visnagar

Gujarat, India

- ▶ Delivered lectures on **Power Systems, Control Systems, Electrical Machines**, and Programming (C, MATLAB) to 500+ undergraduate students
- ▶ Mentored **50+ B.E. project students** through research methodology, implementation, and documentation phases
- ▶ Developed laboratory curriculum including MATLAB simulations, microcontroller programming, and hands-on experiments
- ▶ Organized technical workshops on IoT, embedded systems, and automation technologies

TECHNICAL SKILLS

Dialogflow & CCAI	Dialogflow ES, Dialogflow CX, Conversational Agents (DFCX), CCAI Platform, Agent Assist, CCAI Insights, Voiceflow
LLM & AI Agents	OpenAI GPT-4/GPT-4o, Google Gemini, Claude, LangChain, LlamaIndex, OpenAI Agent SDK, Google ADK, RAG Systems
ML/DL Frameworks	TensorFlow, PyTorch, Keras, scikit-learn, Hugging Face Transformers, dlib, OpenCV, spaCy, NLTK
Cloud Platforms	GCP (Cloud Functions, Cloud Run, BigQuery, Pub/Sub, Vertex AI), AWS (Lambda, EC2, S3, SageMaker)
Backend	Python, FastAPI, Flask, NodeJS, Express.js, TypeScript, REST APIs, GraphQL, Webhook Development
Frontend	React, NextJS 14+, TailwindCSS, ShadCN UI, HTML5, CSS3, JavaScript ES6+
Databases	PostgreSQL, MySQL, MongoDB, Firebase, Elasticsearch, Redis, Pinecone, Weaviate, Qdrant
Integrations	Twilio, WhatsApp API, Messenger, Telegram, Slack, Salesforce, HubSpot, Zendesk, Stripe, Razorpay
DevOps	Docker, Kubernetes, Git, GitHub Actions, Cloud Build, Terraform, CI/CD Pipelines

EDUCATION

M.E. – Power Systems
SPCE Visnagar, Gujarat

CPI: 8.3
2011–2013

B.E. – Electrical Engineering
GIT Jaipur, Rajasthan

67%
2006–2009

KEY ACHIEVEMENTS

- ★ 150+ Production chatbots delivered
- 📺 YouTube channel on Dialogflow/AI
- 🏆 Top-Rated Seller on Fiverr (98%)
- 👥 Mentored 50+ developers/students

🗣️ English, Hindi, Gujarati | ⌚ Available Immediately | 📄 References upon request